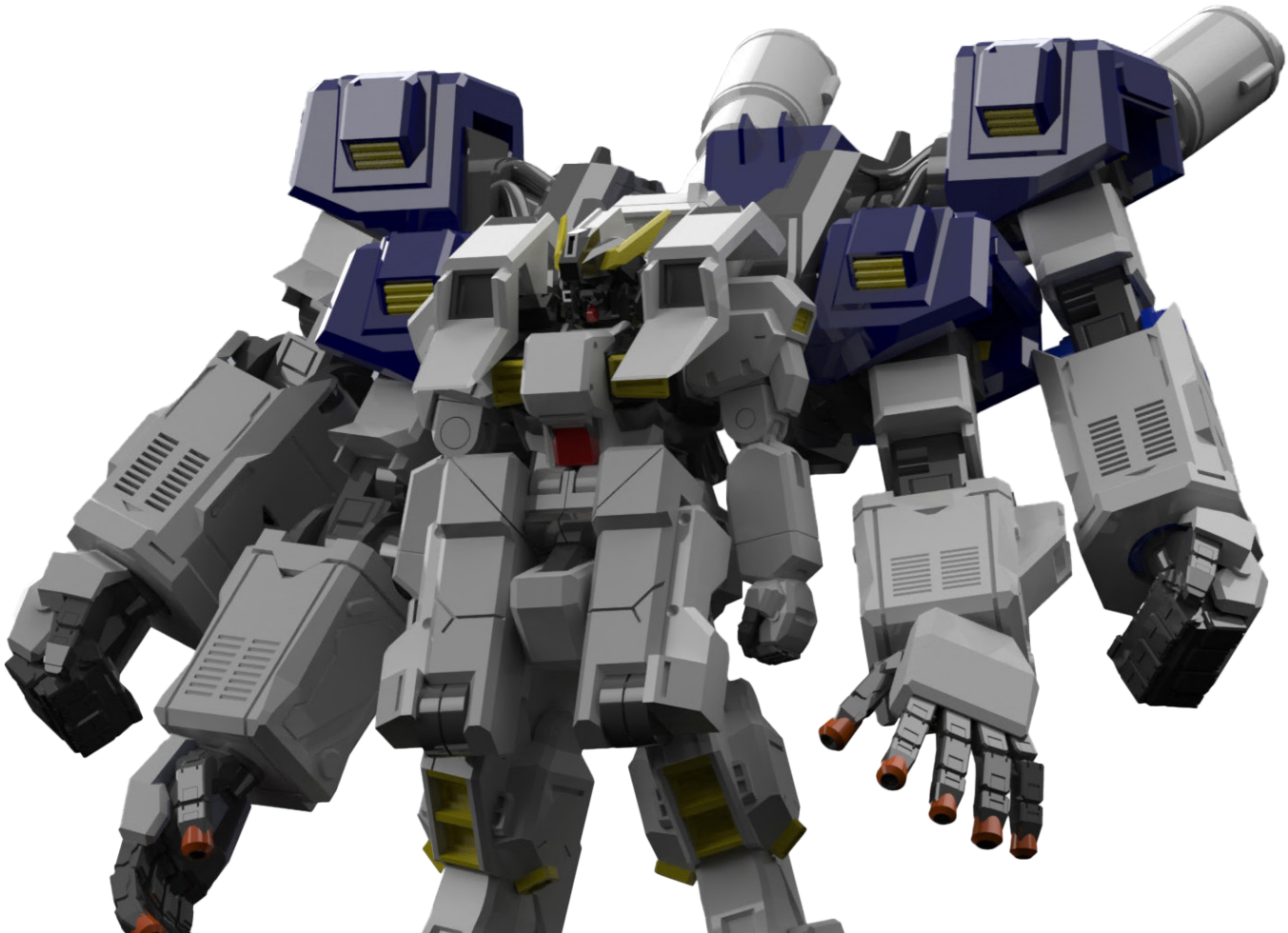
FDM Articulated Figure
RX-121-A²P²/MSPA-711

安装指南 /Installation Guide

For TR-1 Quad Arm[YuA Ver.]



打印前的准备：
Before printing:

试模基于 Bambu PLA Basic 与官方 PEI 板，使用默认配置及 35 度仓温，笔者保证其打印及热床粘接效果，对于其他耗材 / 打印板，请根据实际情况适当调整裙边及其他配置，避免翘边。

1/144 关节零件在首次运动时不要太过用力，左右反复小范围掰动，能够正常运动后就没问题了。

The test was based on Bambu PLA Basic and the official PEI plate, using default setting and a chamber temperature of 35 ℃ . The author guarantees the printing quality and the adhesion to the heated bed. For Other situations, adjust the brim and other settings appropriately according to the actual situation to avoid warping.

Don't use excessive force when moving the 1/144 joint parts for the first time. Gently move them back and forth within a small range . Once they can move normally, 👍.

打印后的处理：
Post printing:

建模时所有连接处均按照理想情况设置间隙，但打印有误差，根据情况切掉一点或塞点胶带。

During the modeling process, the gaps of all joints were set under ideal conditions. However, there are printing problems. You can cut off a little or insert some tape according to the actual situation.

如何寻找零件?
How to find the parts?

你会在 Bambu Studio- 对象一栏看到形如 [X2]objcode.STL 的命名，其中的 X2 便是它在说明书中首次出现的步骤。

You will see names like [X2]objcode.STL in the "Objects" section of Bambu Studio.The "X2" in it represents the step where the part first appears in the instruction manual.

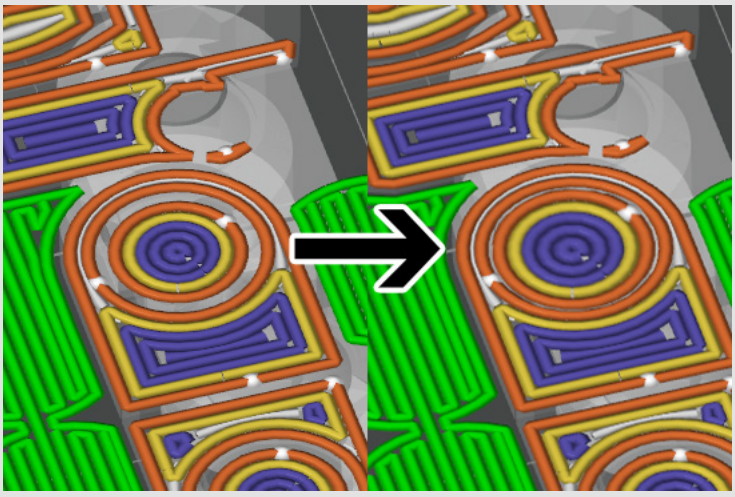
打印质量指南，以关节为例：
Print guide, using joints as sample:

作为精密模型，可动人形系列对打印有要求，我通常使用官方打印机、官方参数、官方耗材试模，而这意味着完全依赖品控，无法复现问题出现的原因自然也就无法解决问题，因此又接触了 klipper 与更多耗材，打印了数十个包含类似关节的零件，总结出一些问题发生的原因。
如果遇到问题，请据此稍做检查。我依然在尽力优化关节系统，争取做到兼顾体积、强度、打印难度。

As precision models, the articulated action figure line has specific requirements for 3D printing. I tested all prototypes with official printers, official print settings, and official filaments, but this left me completely reliant on official quality control. If I could not replicate the root cause of an issue,naturally could not fix it either.So started working with Klipper and testing a wider range of filaments, printed dozens of parts with similar joint designs, and summed up common causes of print issues below. If you run into problems, you can cross-check against this guide. I am still working hard to optimize the joint system to strike a balance between size, strength, and print difficulty.

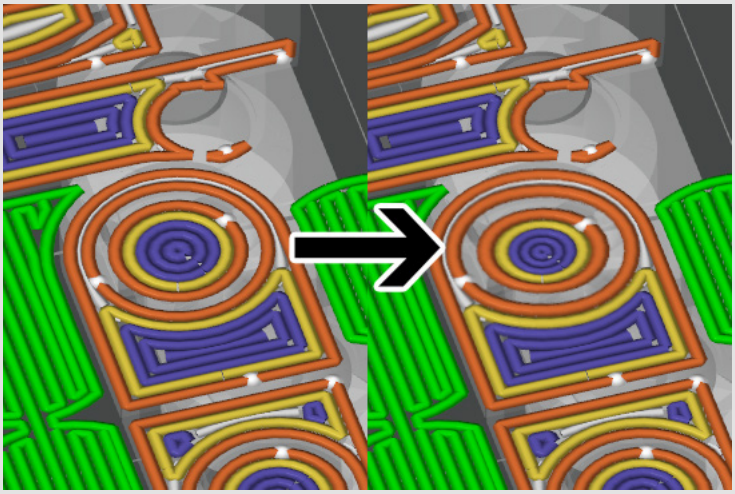
挤出过度：关节融为一体，完全无法移动，会在首次转动时断裂，ABS 居多。

Over-extrusion: Joints fuse into one solid piece and cannot move, and will break the first time you attempt to rotate them. Most common with ABS.



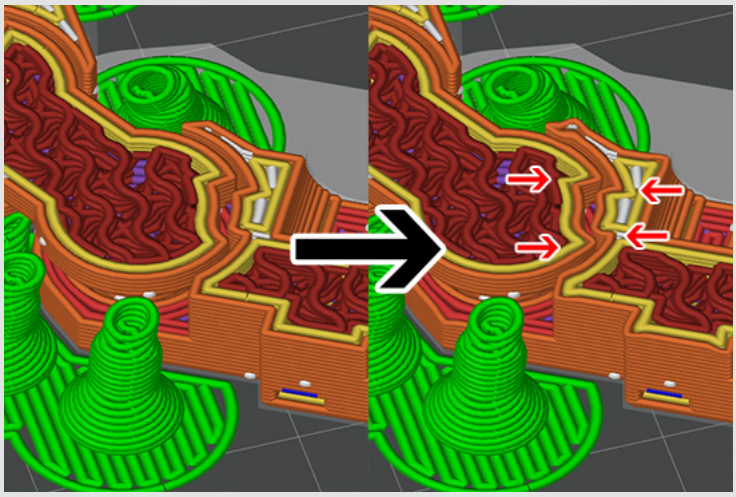
挤出不足：关节空隙偏大，和调平问题同时发生时，零件容易脱离热床。

Under-extrusion: Joints end up with excessively large gaps. When this occurs alongside bed leveling issues, easily detach from the heated bed mid-print.



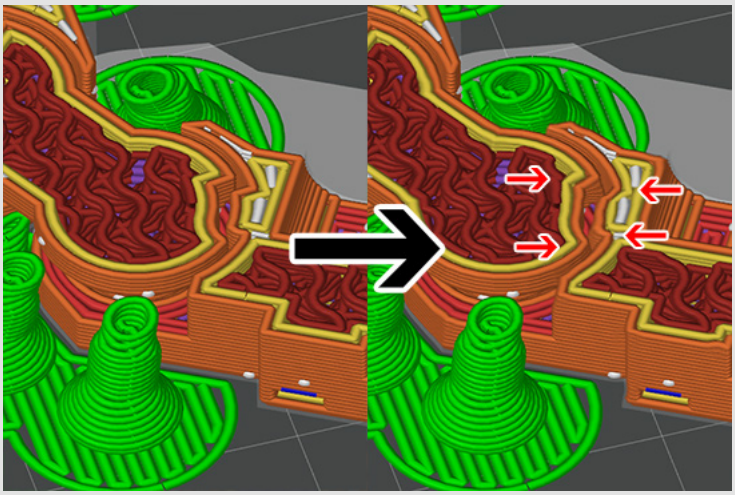
K 值过低: 转角形成多余凸起，关节难以转动，会在首次转动时断裂，多发生在 PETG。

Low K: Excess blobs form at corners, making joints too stiff to turn. They will usually break on the first rotation, most often with PETG.



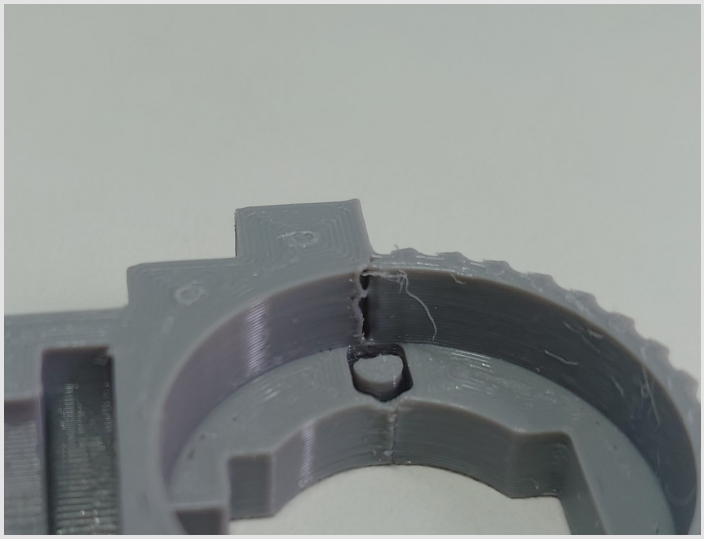
K 值过高：转角非常圆润，对于段落关节，会感到手感非常顺滑，然后站不起来。

High K: Corners come out overly rounded. For segmented joints, this will feel very smooth when operated, model will be unable to stand.



接缝问题：形成或凸起或凹陷的缝隙，通常与冷却与 K 值相关。不过只要维持随机接缝设置，让凹凸无法积累，一般不会带来问题。

Seam: Large noticeable gaps (raised or recessed) form at seams. Uusually related to cooling settings and K. However,



keep the random seam setting enabled to prevent unevenness from accumulating, this rarely causes any functional problems.

温度问题：更低的强度，运动时从层纹断裂，悬垂转角处会出现下垂。

Temperature: Results in reduced part strength. Layers will split apart along layer lines during movement, and drooping will occur on overhanging corners.

热床不粘：我不知道为什么 … 同样的零件，有时候会听到掉下热床的反馈，我却要为取下零件而头疼，因为经常用力过猛而伤着手，左手拇指的疤痕至今依然鲜红 …

Poor bed adhesion: I can’t even figure this one out… For the exact same part, I hear other users report their part falling off the Hbed mid-print, but for me: I always struggle to pry the finished part off the Hbed, and cut my hand multiple times from pushing too hard. The scar on my left thumb is still bright red to this day…

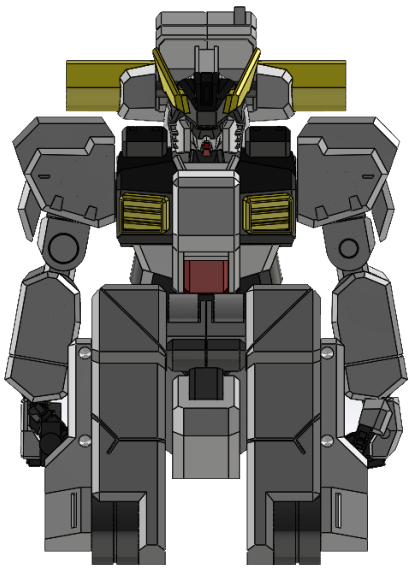
出现在低价耗材上的未知问题：阶梯一样的凸起，通常认为是回抽与温度共同结果，即流动性与预期不符，但反复调整温度、回抽、转角、冷却、速度、加速度、K 值等等依然无法取得最好的效果，把它用在拓竹打印机上，问题减轻却依然存在。如果这种情况发生在关节内部，关节将被卡死。

Unknow issues with cheap filaments: Generally assumed to be caused by a combination of retraction and temperature settings, meaning filament fluidity does not match expectations. But even after repeatedly adjusting everything else, still cannot get a perfect result. Printing these filaments on bambu printers reduces the issue but does not eliminate it entirely. If this problem occurs inside a joint, the joint will get completely stuck.

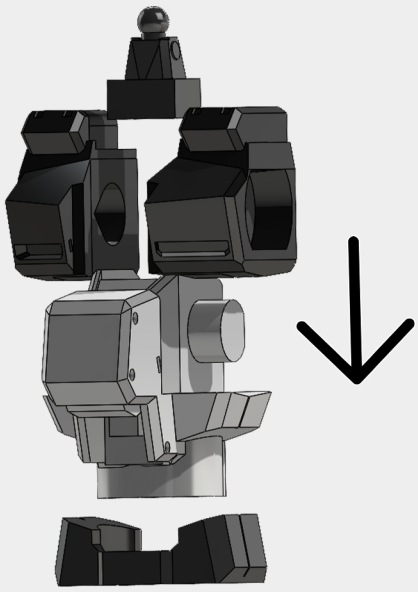
我也尝试过令耗材充分受潮，它没有带来任何问题，表面不佳只会影响部分关节的顺滑度，此时滴入润滑油可以解决。

I have also intentionally tested printing with fully moisture-soaked filament, and it does not cause any major functional issues. only the poorer surface finish reduces the smoothness of some joints, which can be easily fixed by adding a drop of lubricating oil.

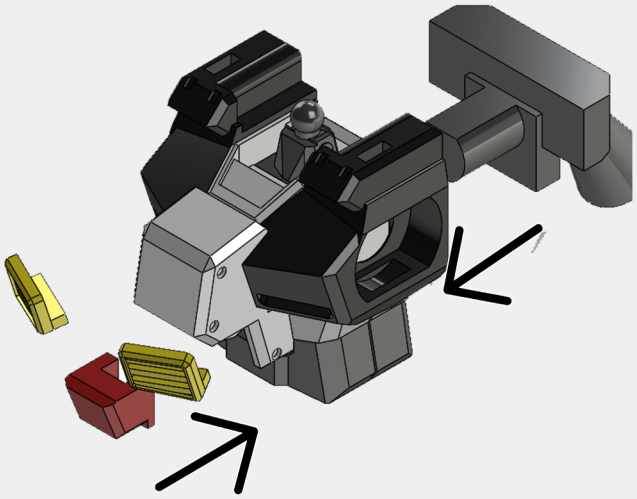
上半身 /Upper body



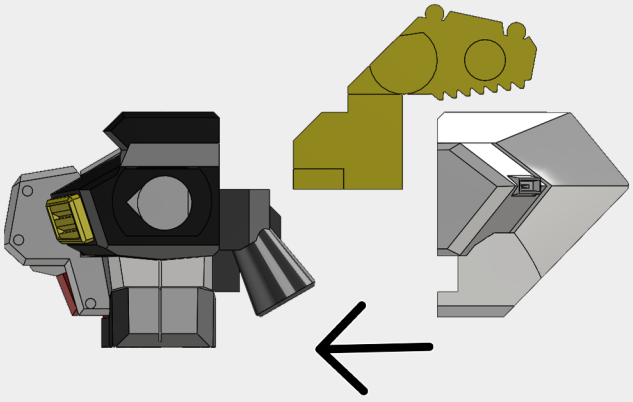
A1



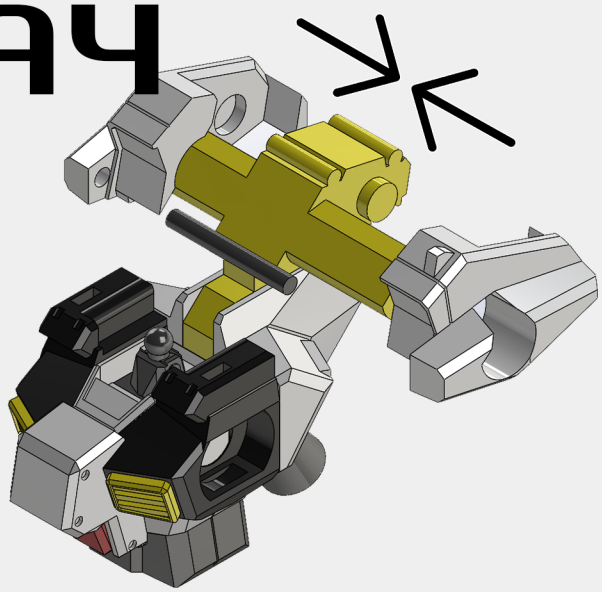
A2



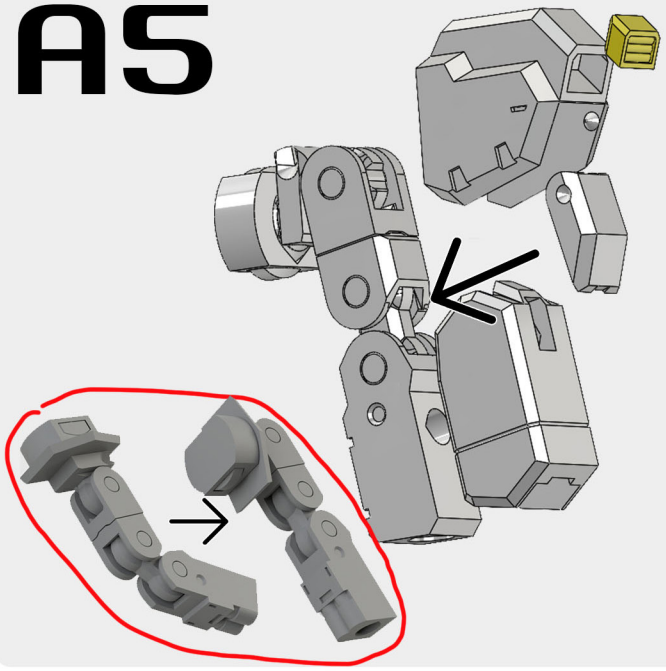
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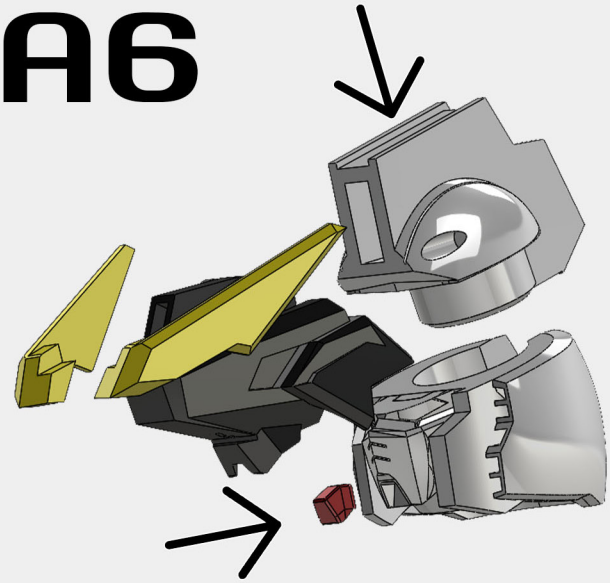
A4



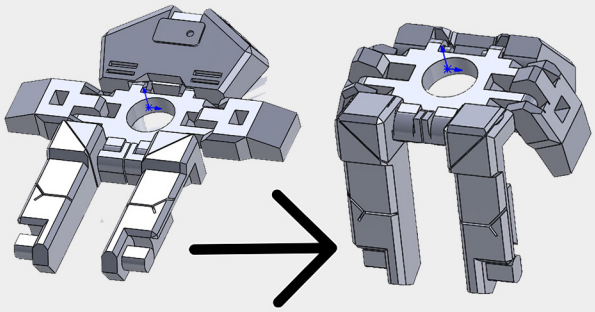
A5



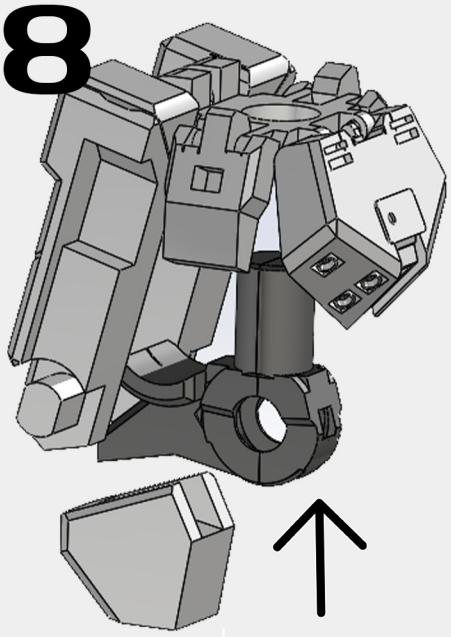
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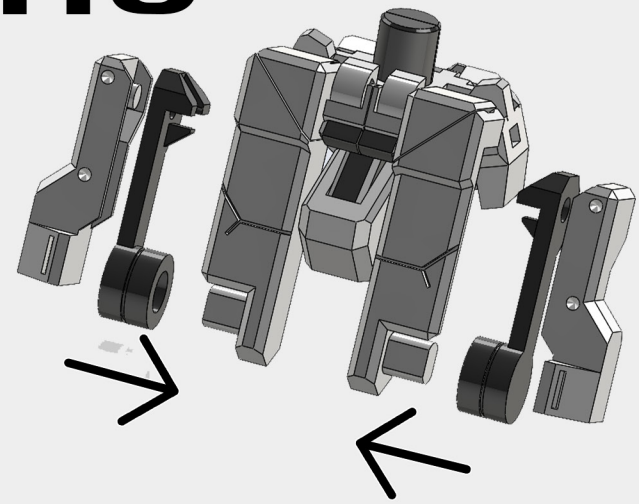
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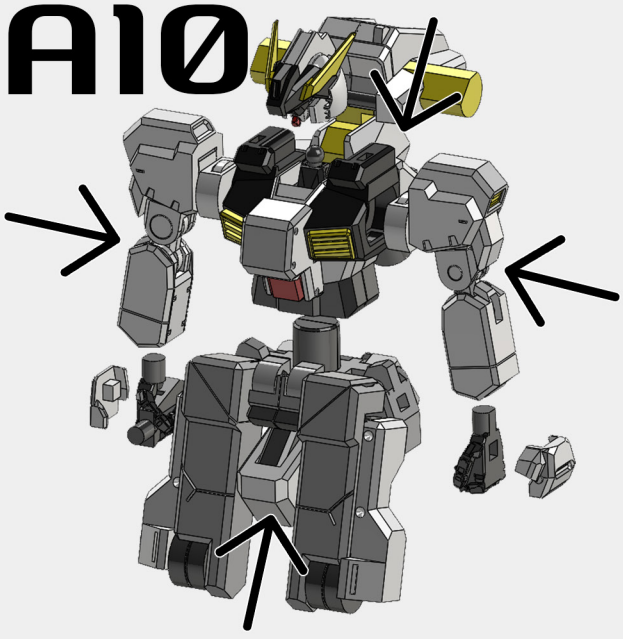
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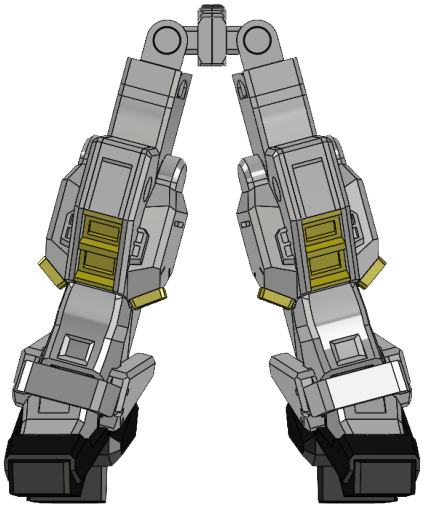
A9



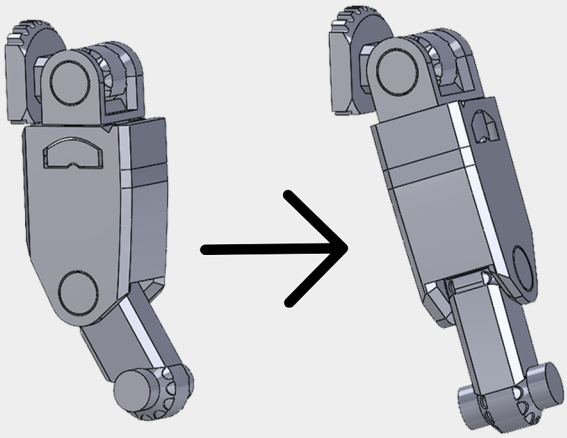
A10



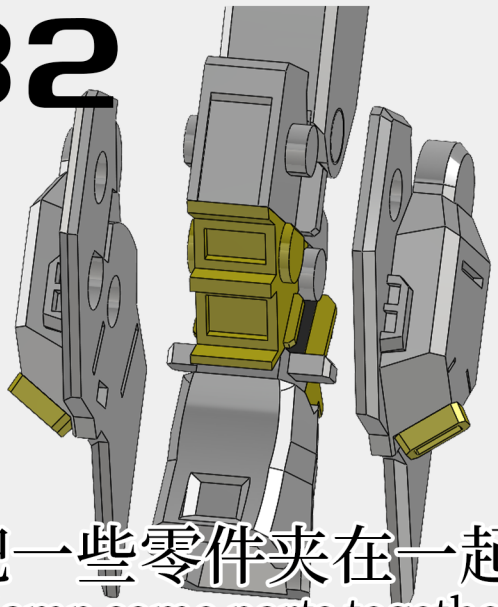
腿部 /LEG



B1



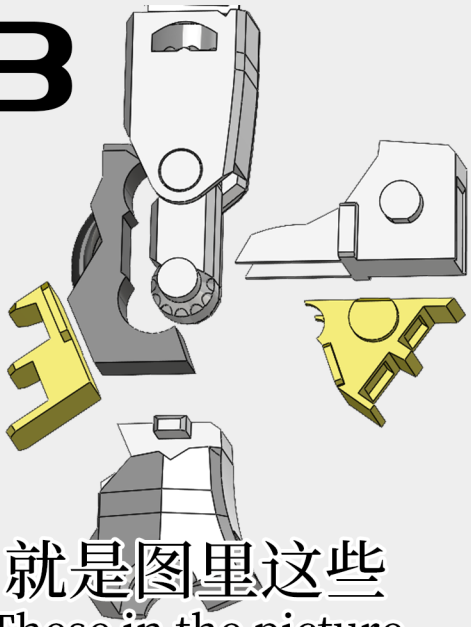
B2



把一些零件夹在一起

Clamp some parts together

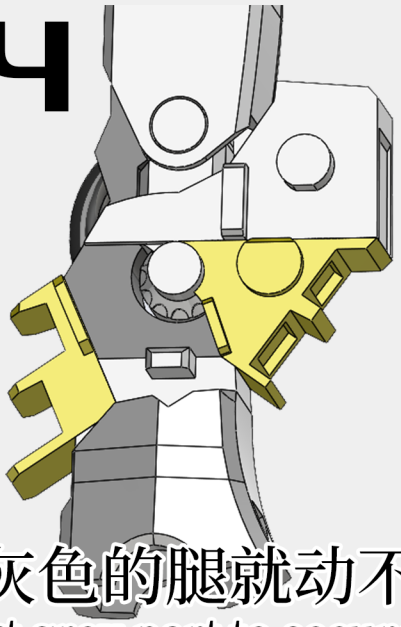
B3



就是图里这些

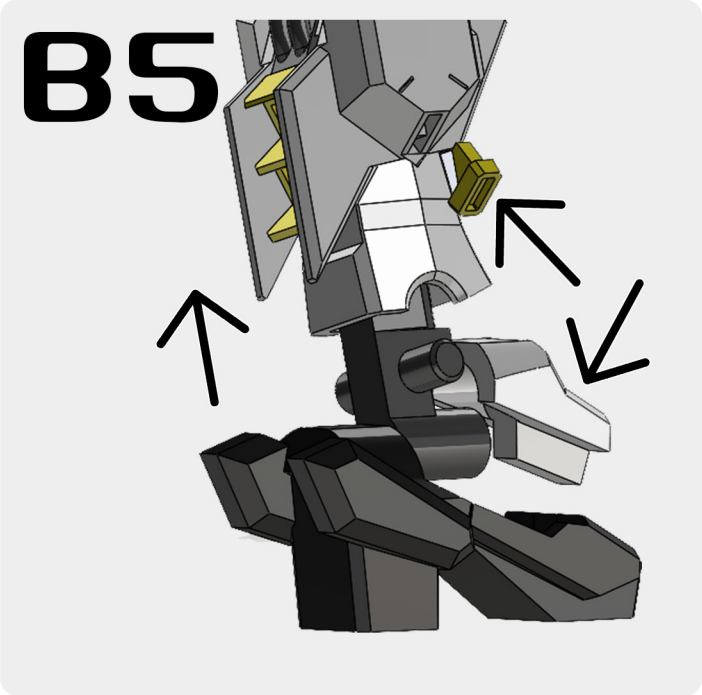
These in the picture

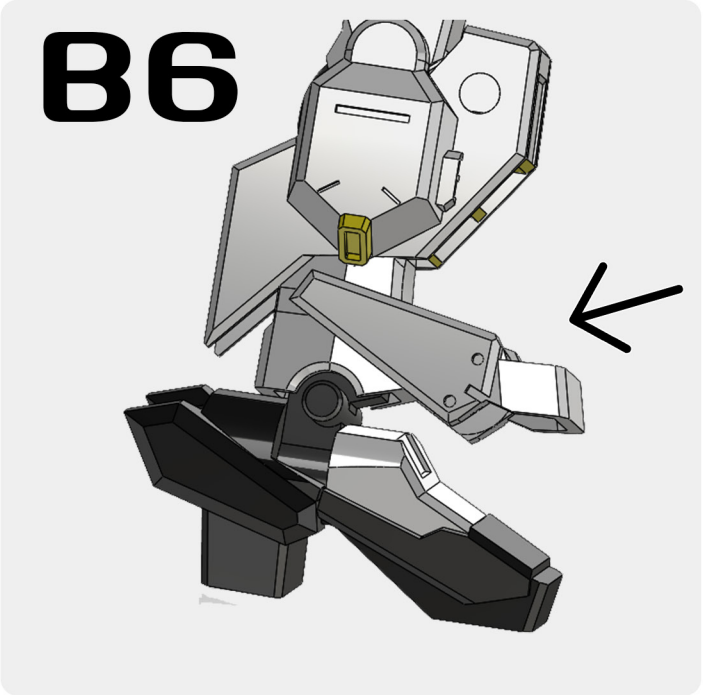
B4

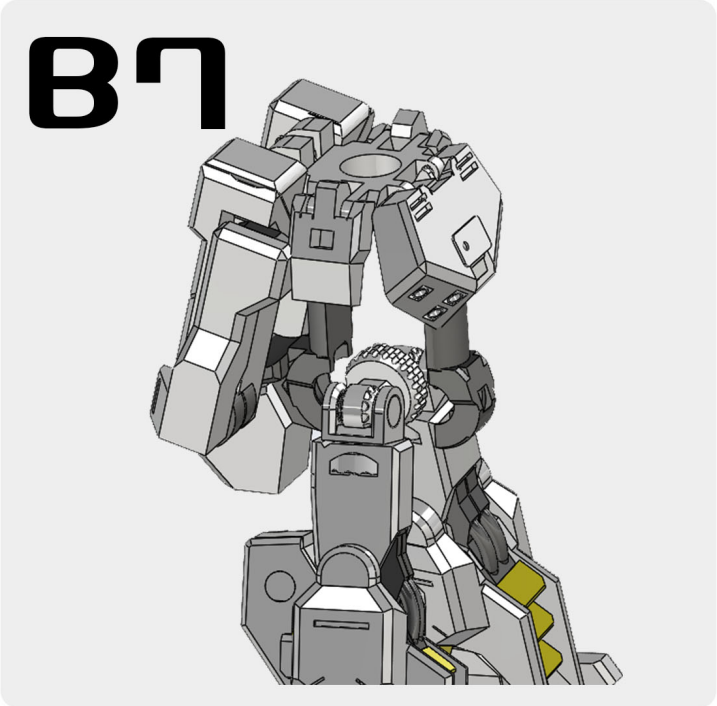


装了灰色的腿就动不了了

Insert grey part to secure leg







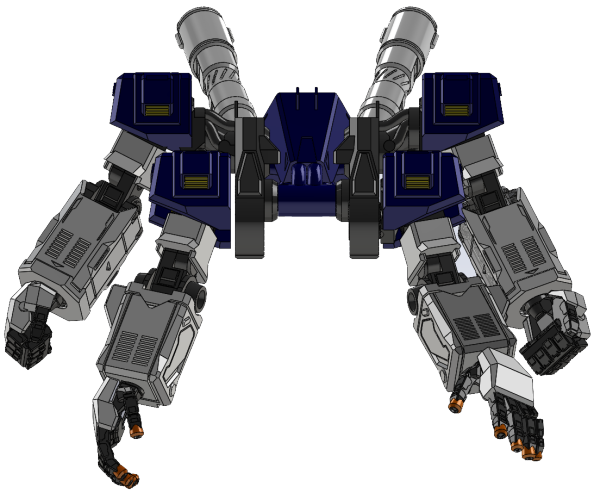
其实应该先装腿，

然后涂点啫喱胶可以

避免骨盆前倾。

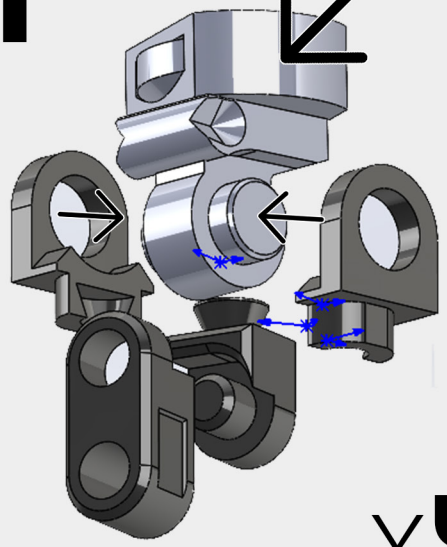
Actually, we should put this first, apply a little gel can help prevent a...n anterior pelvic tilt.

背包 /BACKPACK



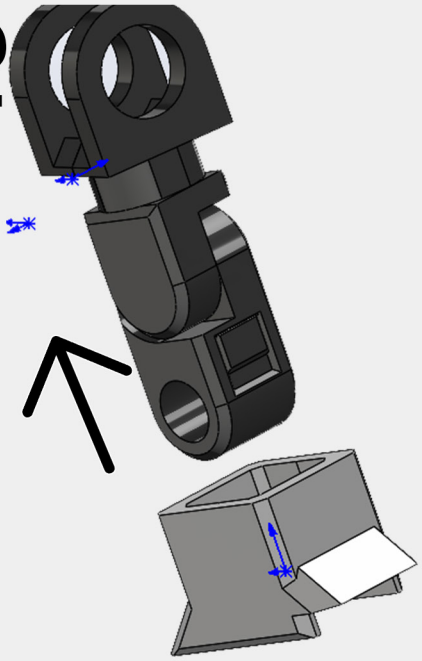
C1

2 TYPE

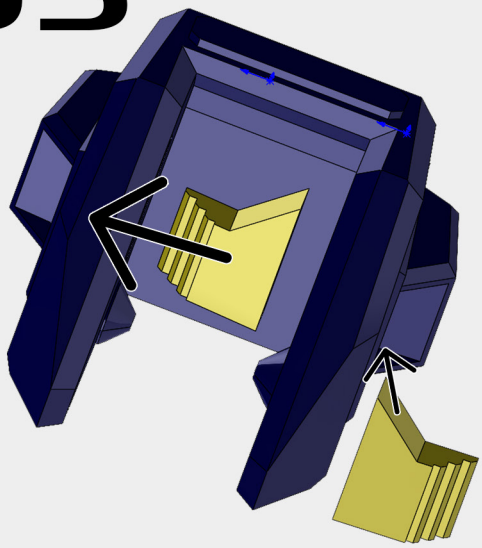


X4

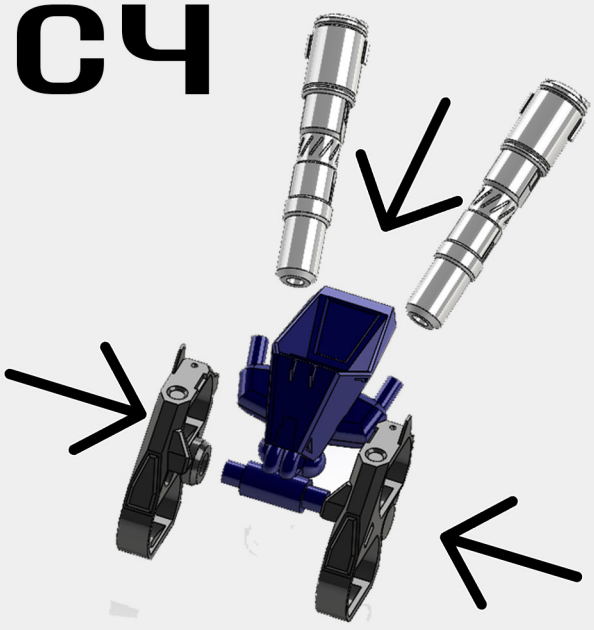
C2



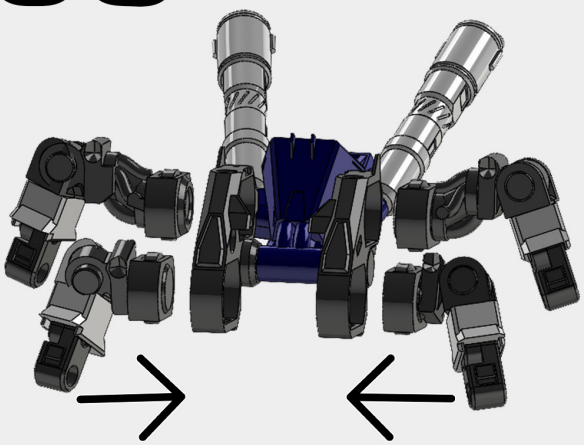
C3



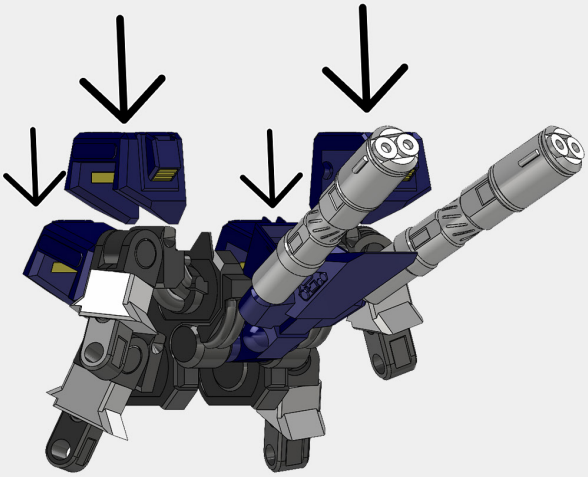
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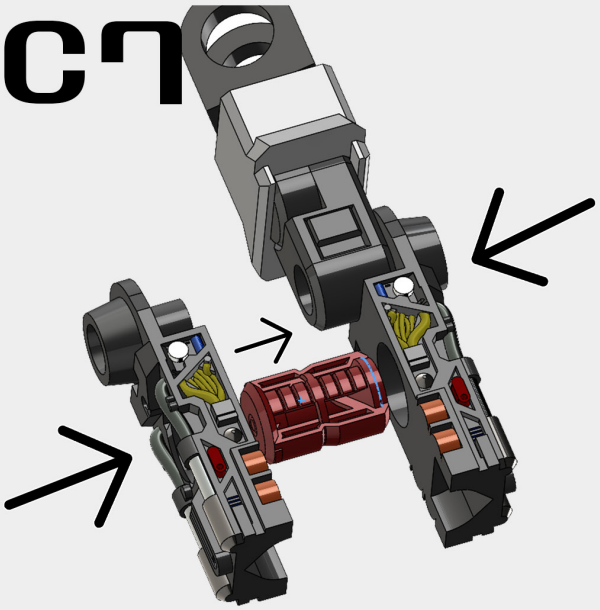
C5



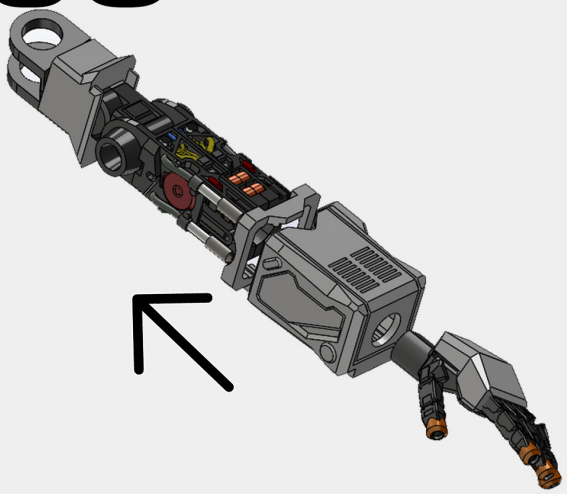
C6



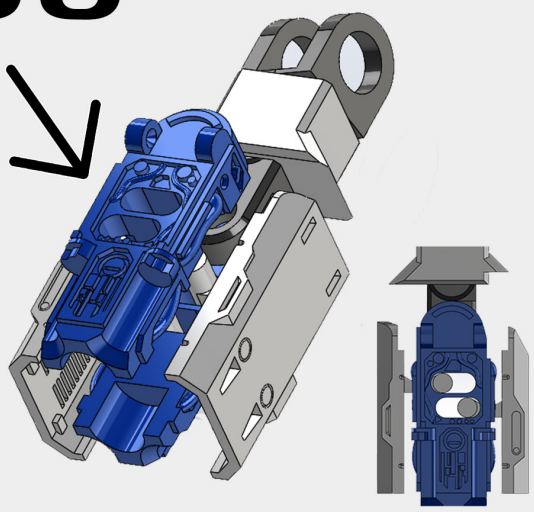
C7



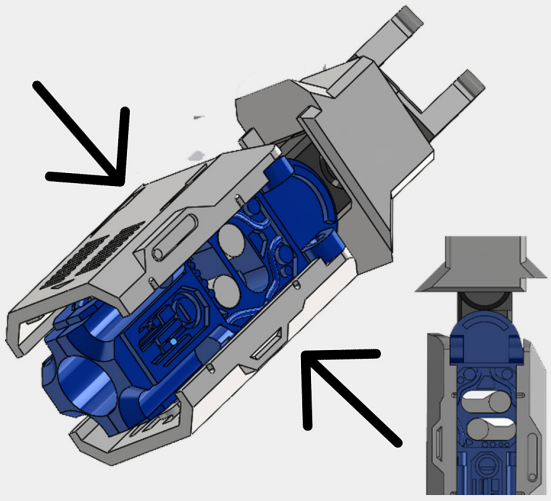
C8



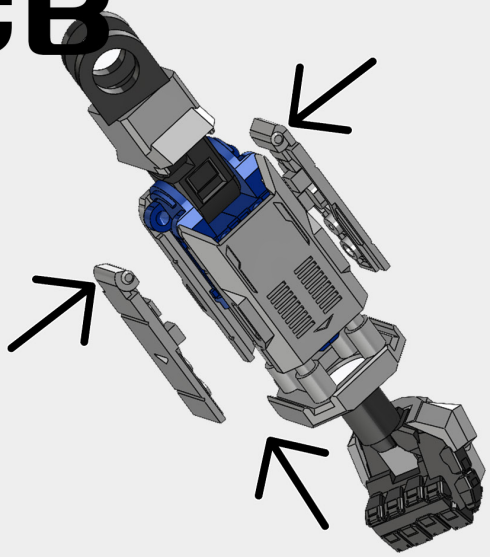
C9



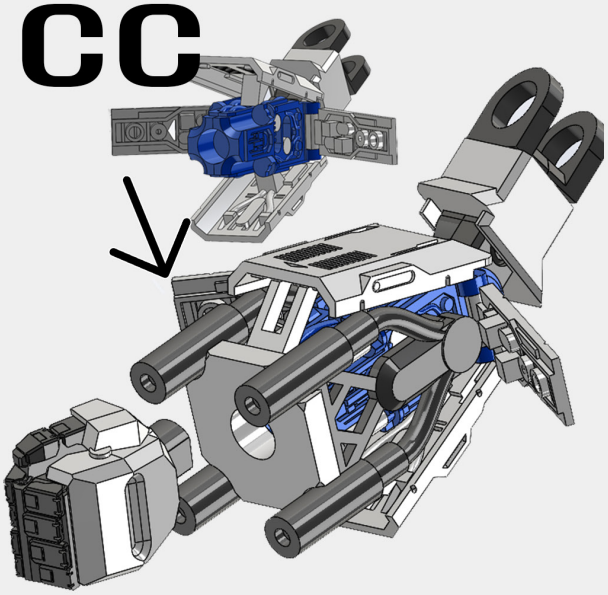
CA



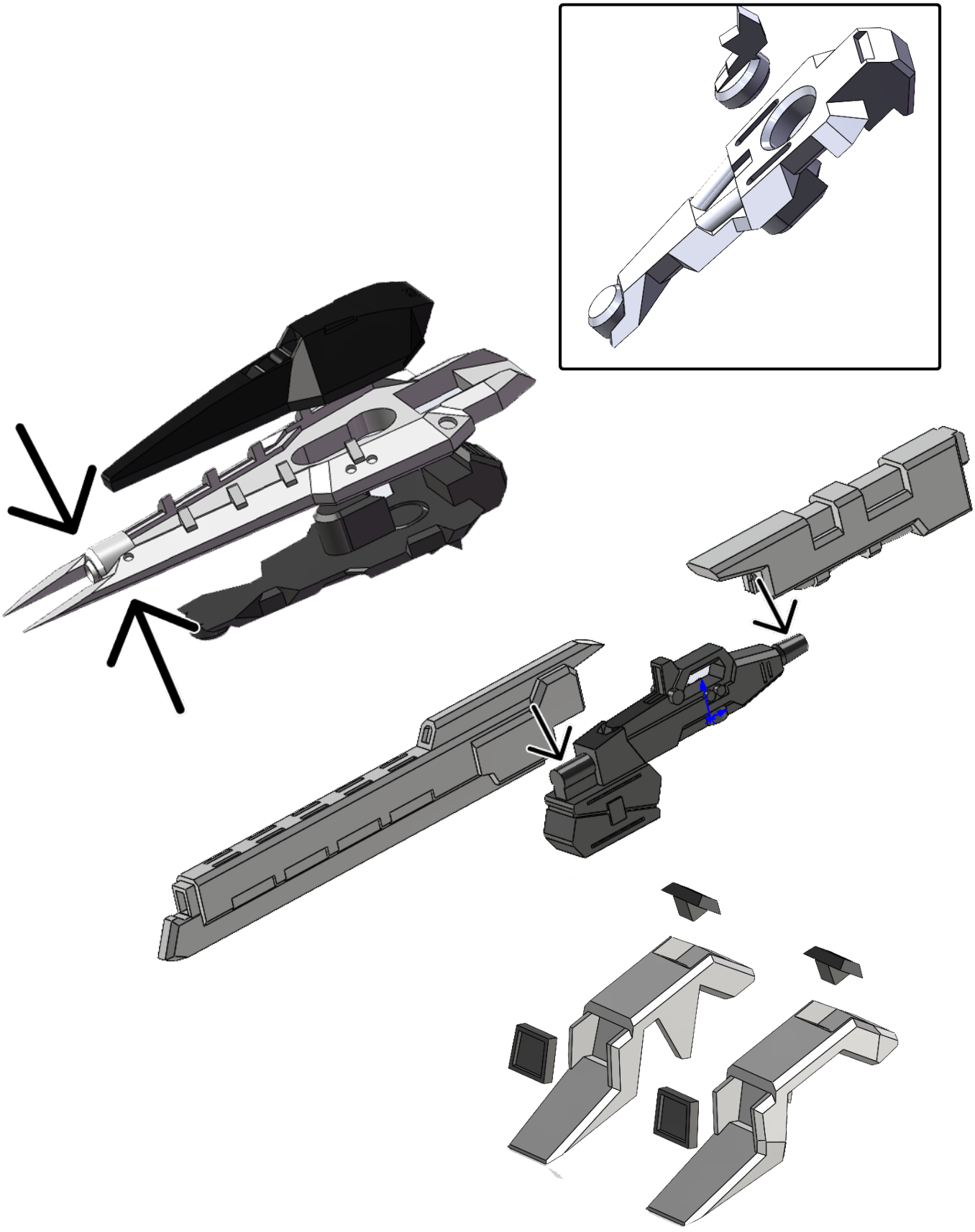
CB



CC



武器 /WPN



宣传图 Poster


FDM Articulated Figure
TR-1 Quad-Arm
RX-121-A²P²/MSPA-711


独立本体与多重展开机构
/Multi-layered Structure


武器系统兼容LUCF
/Compatible with LUCF


良好结构强度与稳定性
/High Strength&Stability